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Math 300-902

Topic: Descartes and Fermat Pre-Calculus Feud (Custom Topic)

Title: The Legacy of the Descartes-Fermat Feud on Analytic Geometry and Calculus

Abstract:

This term paper analyzes the feud between French mathematicians Pierre de Fermat and René Descartes over the discovery of a tangent line of a curve. This paper also highlights the limitations of Descartes' method and Fermat's informal style of writing proofs. The feud between Fermat and Descartes helped form the basic intuitions of calculus that would be picked up by Isaac Newton and Gottfried Leibniz for the foundations of Calculus. Overall, the paper offers insight into the early developments of analytic geometry and how the feud between Fermat and Descartes helped shape the course of calculus.

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Introduction

The discovery for the computation of the tangent line to a curve has been a fundamental problem in mathematics for centuries. The development of the concept of a tangent line to a curve started with early Greek ideas of analytic geometry from the works of Euclid of Alexandria and Apollonius of Perga in 300 BCE and 225 BCE respectively. In the 17th century, French mathematicians Rene Descartes and Pierre de Fermat significantly contributed to finding the tangent line to a curve. Their methods were different, and they did not see eye to eye on how the other came to their respective conclusions.

However, this feud helped form the basic intuitions of calculus that would be picked up by Newton and Leibniz for the foundations of Calculus. This paper aims to analyze the feud between Fermat and Descartes over the discovery of computing the tangent line of a curve. This paper will also provide background information on the origins of the tangent line, explain the tangent line problem, analyze Descartes' and Fermat's methods for finding the tangent line, and evaluate the significance of the feud to the development of calculus.

Section 1: Greeks and the Genesis of the tangent line

The origins of the tangent line date back to early Greek ideas of analytic geometry. In their respective works, Elements, and Conics, Euclid and Apollonius had similar definitions of tangent lines. Although their works spoke of tangents to circles, these works paved the way for mathematical ideas going forward and would be picked up by French mathematicians Fermat and Descartes in the 17th Century.

In Euclid's "Elements" Book 3, Proposition 18, he defined a tangent line to a circle as a line that touches the circle at only one point. He also showed that the tangent line is perpendicular to the radius that goes through the point of tangency. Apollonius, in his work "Conics" Book 1, Proposition 17, also defined a tangent line to a conic section as a line that intersects the curve at only one point.

These early ideas of the tangent line were limited to curves that were explicitly defined as circles or conic sections. However, they laid the groundwork for later mathematicians to generalize the concept of the tangent line to more complex curves and to develop methods for finding the tangent line at any point on a curve.

Section 2: Importance of the Tangent line problem

Imagine a time before calculus, without the knowledge of power rule or the concept of limits. For any curve that does not have a constant rate of change, there was no simple way to compute the said rate of change at any point on the given curve. This scenario was a reality for mathematicians prior to Calculus and eventually culminated in the issue of the Tangent Line Problem. The tangent line problem is an important concept in mathematics with numerous practical applications. It is used to determine the instantaneous rate of change of a function at a particular point, which is crucial in many fields of science like engineering, and economics.

For example, in physics, the tangent line problem can be used to determine the instantaneous velocity of an object at a specific moment in time. This is done by finding the slope of the tangent line to the object's position function at that point. The same concept can be

applied to acceleration, which can be found by taking the derivative of velocity with respect to time. In engineering, the tangent line problem is used to find the rate of change of various physical quantities, such as temperature, pressure, and flow rate. This is important in designing and optimizing various systems, such as heating and cooling systems, fluid dynamics, and combustion engines. In economics, the tangent line problem is used to find the marginal cost and revenue functions, which help businesses determine the optimal price and production levels to maximize profits. The marginal cost is the instantaneous rate of change of the total cost function, while the marginal revenue is the instantaneous rate of change of the total revenue function.

Overall, the tangent line problem is a fundamental concept that plays a crucial role in many areas of mathematics and its applications. It is a powerful tool that allows us to understand the behavior of functions at specific points and determine important rates of change that are necessary for solving many real-world problems.

Section 3: Rene Descartes and Computing the Tangent Line of a Curve via the Method of Normals

Rene Descartes was a French philosopher and respected mathematician of the 16th century. Descartes formalized ideas that related geometric shapes and algebraic techniques with the development of the Cartesian coordinate system in his work, La Géométrie, which united algebra and geometry into a unified subject named Analytic Geometry. This innovation revolutionized how curves are analyzed to this day. With a standardized way to study curves, Descartes then sought to solve the question of computing the tangent line of a curve.

Rene Descartes developed a method for finding the tangent line to a curve at a given point, known as the Method of Normals. The method involves finding the line perpendicular to

the tangent line at the point of interest, called the normal line. This tangent equation would also be tangent to some arbitrary circle with a radius length of the original point to the x-axis in a straight line. Then with numerous algebraic manipulations of these found values, and applying the Factor Theorem, eventually the slope of the normal line was found, which happened to be the negative reciprocal of the slope of the original line. Reference Figure 1 for a detailed example and informal proof of this process.

This method was very plausible and definitely required some ingenuity but was overly complex for curves represented by polynomials. Also, this method required the curve in question to be explicitly defined. In other words, implicitly defined functions were, at the time, seemingly not possible to have a method for finding a tangent line at a given point.

Section 4: Pierre De Fermat and Computing the Tangent Line of a Curve via Adequality

Pierre De Fermat was a French judge and hobbyist mathematician of the 17th century. Perhaps known most for the theorem, Fermat's Last Theorem, Fermat also independently discovered a way to compute the tangent line to a curve at any point. Alongside other mathematicians of the time, Fermat made efforts in the field of Analytic Geometry.

Fermat developed his method for finding the tangent line to a curve using the concept of adequality, which is a notion of equality between infinitesimal quantities. Fermat's approach first involved redefining the function into a multivariable function of x and y and then comparing two infinitesimal quantities associated with the curve at the point of interest, one corresponding to a small change in the value of x and the other corresponding to the corresponding change in

the value of y. By setting these two infinitesimal quantities equal, and canceling, Fermat was able to find the slope of the tangent line at the point of interest. This gives the illusion of comparing two points similar to finding the secant line between the two points, but eventually it would be determined the difference between them is negligible. The process of Adequality can be seen in Figure 2 located in the appendix.

This method proved to be quite ingenious as well. The ability to find the tangent line for implicit functions, allows for complex functions that may rely on more than one independent variable to be evaluated. A common example of a function that would now have a computable tangent line function would be the logarithmic function. Commonly denoted $log_b y = x$. This function is implicitly defined and Fermat's Method of Adequality allows for a tangent to be found.

To the calculus-inclined individual, this method does intuitively seem to be incredibly similar to the concept of limits and the limit definition of the derivative. While this method was seemingly simpler than Descartes, it initially was not taken seriously by Descartes, who even accredited his own work as the reason for Fermat's conclusions. As well, whenever publishing works, Fermat was infamous for leaving some steps in his proofs ambiguous, which left the professional mathematics community to lean more towards Descartes' method initially.

Section 5: Analyzing the Feud and the Birth of Calculus

The rivalry between René Descartes and Pierre de Fermat was long-standing and often believed to be a contentious one. The two mathematicians had differing ideas about the nature of

the mathematical inquiry, and their disagreements over the tangent line to a curve were just one manifestation of this broader divide.

Descartes believed that Fermat's method was not rigorous and relied too heavily on intuitive reasoning. Descartes also criticized Fermat's use of infinitesimals, which were not well-defined at the time. Fermat's usage of infinitesimals was deemed "hand-wavey" by the mathematical community as during his application, there seemed to be a point where a term was specifically canceled out of both sides of an equation and then subsequently set to zero. This specific process was deemed too close to dividing by zero, which has undefined behavior. In hindsight, this method was very close to the calculus idea to limits. Although Fermat's method was more general than Descartes' method, and could be applied to implicitly defined functions as well, it would undergo scrutiny for its method in practice. This would culminate into a feud between the two mathematicians as both believed their methods were superior.

This feud was then turned into a challenge from Descartes to Fermat when Descartes challenged Fermat to find the tangent line of a specific curve (G. F. Simmons 101). The famous curve, Folium of Descartes, was proposed to have no way to calculate the tangent line at any given point Descartes as it was implicitly defined: $x^3 + y^3 + 3dxy$, where d is simply a constant. The goal of this challenge was to show Fermat's method was inferior to his own. Descartes was then proven wrong by Fermat as he indeed was able to calculate the tangent of the Folium of Descartes at a specific point and as a result, Fermat was given more notoriety for his works. An example of Fermat's method can be shown in Figure 3 for a demonstration involving the Folium of Descartes.

Given the nature of Fermat's method of adequality involving infinitesimally small quantities and the hindsight of present-day Calculus, it can be assumed this way of thinking led to the discovery of Calculus. Additionally, the feud between Descartes and Fermat helped to popularize the study of analytic geometry. These developments opened the door for future mathematicians, such as Isaac Newton and Gottfried Leibniz, to develop calculus as a formal mathematical system for dealing with rates of change and slopes of curves.

Newton and Leibniz developed the concept of Calculus independently of one another in the mid to late 17th Century. Just as Fermat and Descartes feuded over the correct way to solve for tangents of curves, Newton and Leibniz feuded over who was to be credited for the discovery of Calculus. The two mathematicians had different notations and different methods for solving newly formed calculus-based problems which only further fueled the rivalry. In the end, Newton was recognized as the person who discovered Calculus while Leibniz's notation was used more widely within the professional mathematics community. This discovery was directly used to innovate the world of natural science, now called Physics.

Conclusion

In conclusion, the feud between René Descartes and Pierre de Fermat over their respective methods for finding the tangent line to a curve highlights the important role of mathematical controversy and debate in the development of mathematics. While both mathematicians made significant contributions to the field of calculus, they approached the problem of finding the tangent line in different ways, with Descartes favoring the method of normals and Fermat relying on adequality and similar triangles.

Ultimately, the feud between Descartes and Fermat serves as a reminder of the importance of debate and disagreement in the pursuit of mathematical truth, and the vital role that collaboration and competition play in advancing our understanding of the world around us.

Appendix

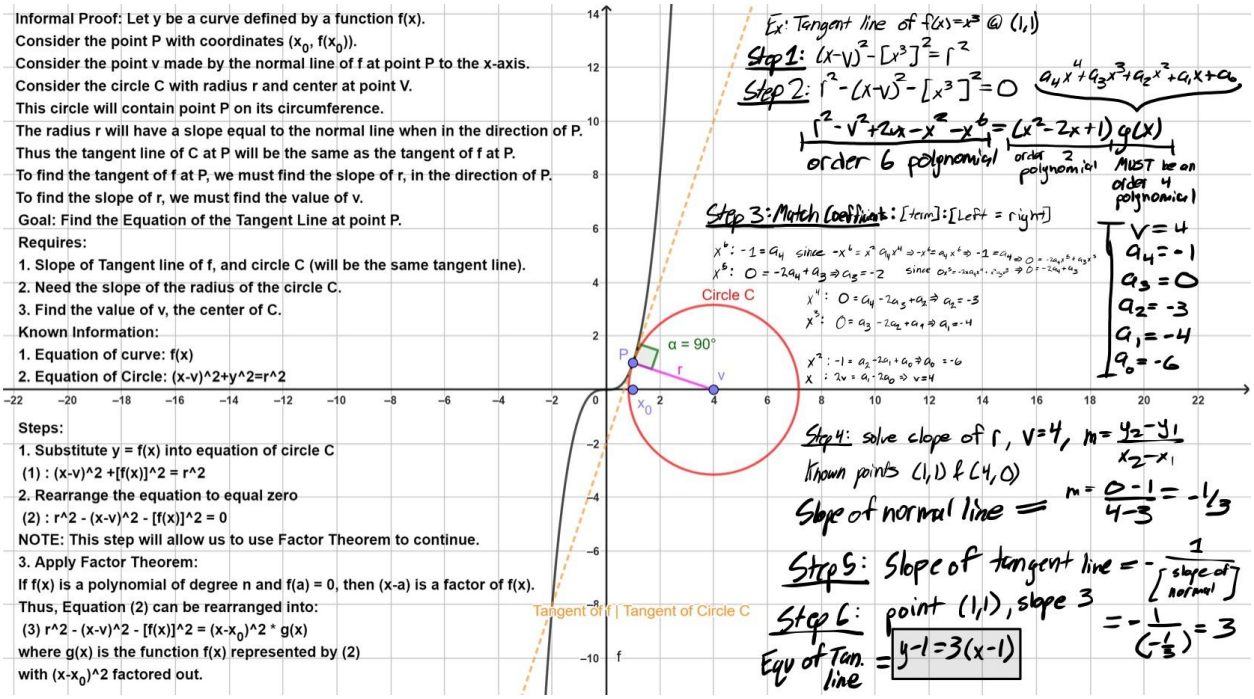


Figure 1: Image of Descartes method of normals for a general curve f(x) and applying it to the curve $f(x) = x^3$.

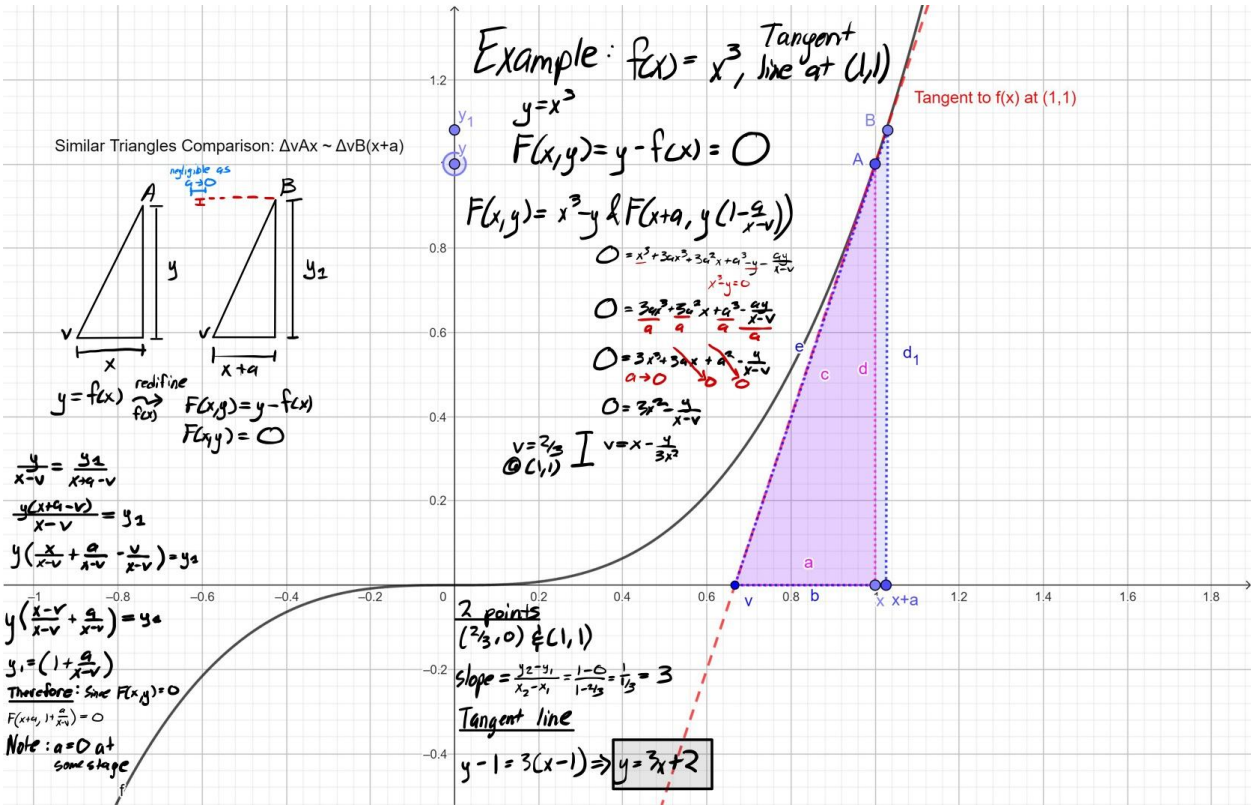


Figure 2: Image showing a completed example of Fermat's method of adequality to find the tangent line of a curve at a point with the curve $f(x) = x^3$

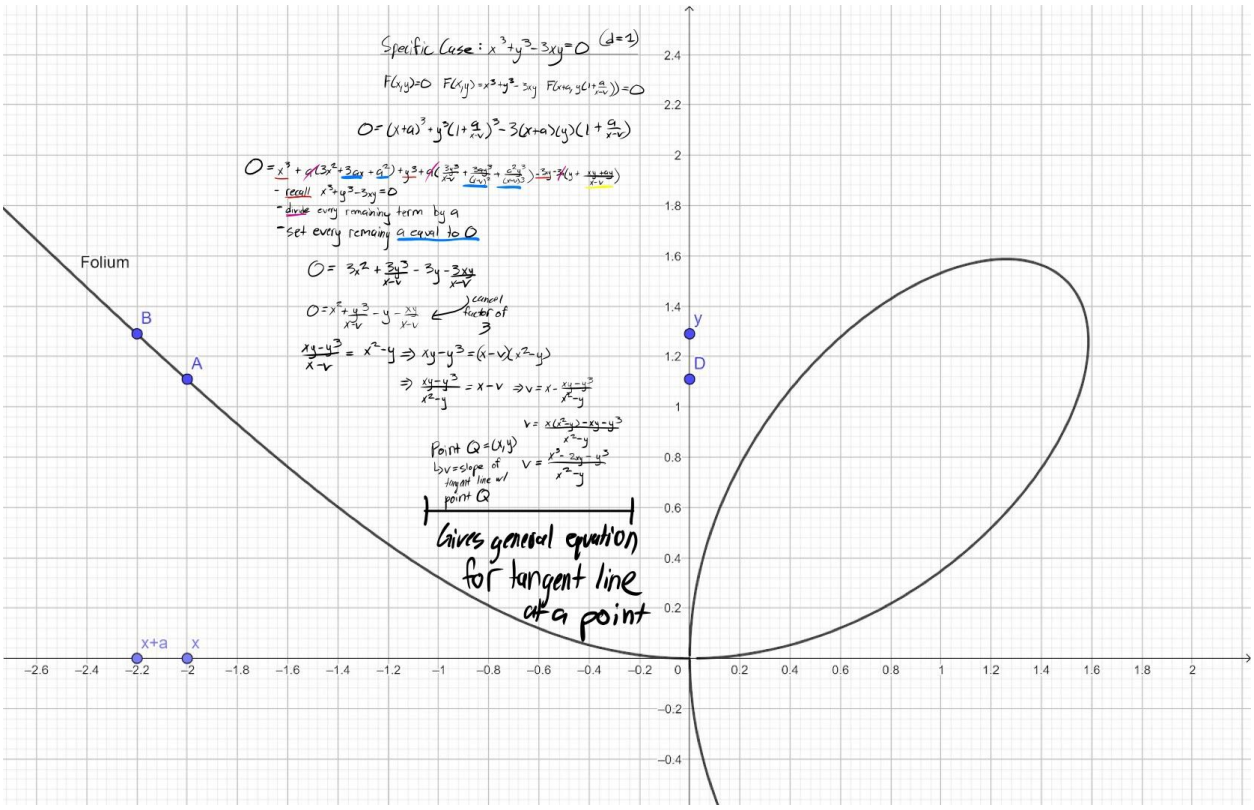


Figure 3: Image showing Fermat's method solving the tangent on the Folium of Descartes, an implicit function defined $x^3 + y^3 - 3dxy = 0$, where d is a constant.

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